Computational Gastronomy
Prof. Ganesh Bagler

Monsoon 2026 – Week 01/02

- **Some broad questions that we shall touch upon:**
  - What is food?
  - What is technology?
  - What are the ways in which food touches are lives?

- **Ingredient Categories: Form & Shapes**
  Vegetable: All kind of leafy vegetables (spinach, fenugreek), beans (kidney bean, red beans), tubers (potato, turnip, beetroot) and such.
  Spice: Flavorful ingredients primarily from root, bark and seeds. Examples: cumin, clove, cardamom, cinnamon, turmeric etc.
  Herb: Flavorful ingredients primarily from leaves and stems of plants. Examples: coriander, thyme, rosemary, mint, basil etc.
  Pulse: The fruit or seed of such. Examples: chickpeas, green gram, back gram, pigeon pea etc.
  Cereal/Crop: Examples: rice, oat, sorghum, barley etc.
  Plant: Examples: tea, thistle, pine, hop
  Fruit: Examples: apple, orange, guava, grape, papaya etc.
  Flower: rose, jasmine, lavender, artichoke etc.
  Nut/Seed: Examples: peanut, walnut, hazelnut, filbert etc.
  Dairy: Example: milk, curd/yogurt, butter, buttermilk, cream, cheese, whey etc.
  Meat: Examples: beef, pork, mutton, chicken etc.
  Fish: Examples: salmon, codfish, tuna, catfish, bluefish etc.
  Seafood: Examples: lobster, mollusk, oyster, prawn, shrimp, kombu, kelp etc.
  Beverage-Alcoholic: Examples: rum, gin, wine, whisky etc.
  Beverage: fruit juice, soft drink, milk shake etc.

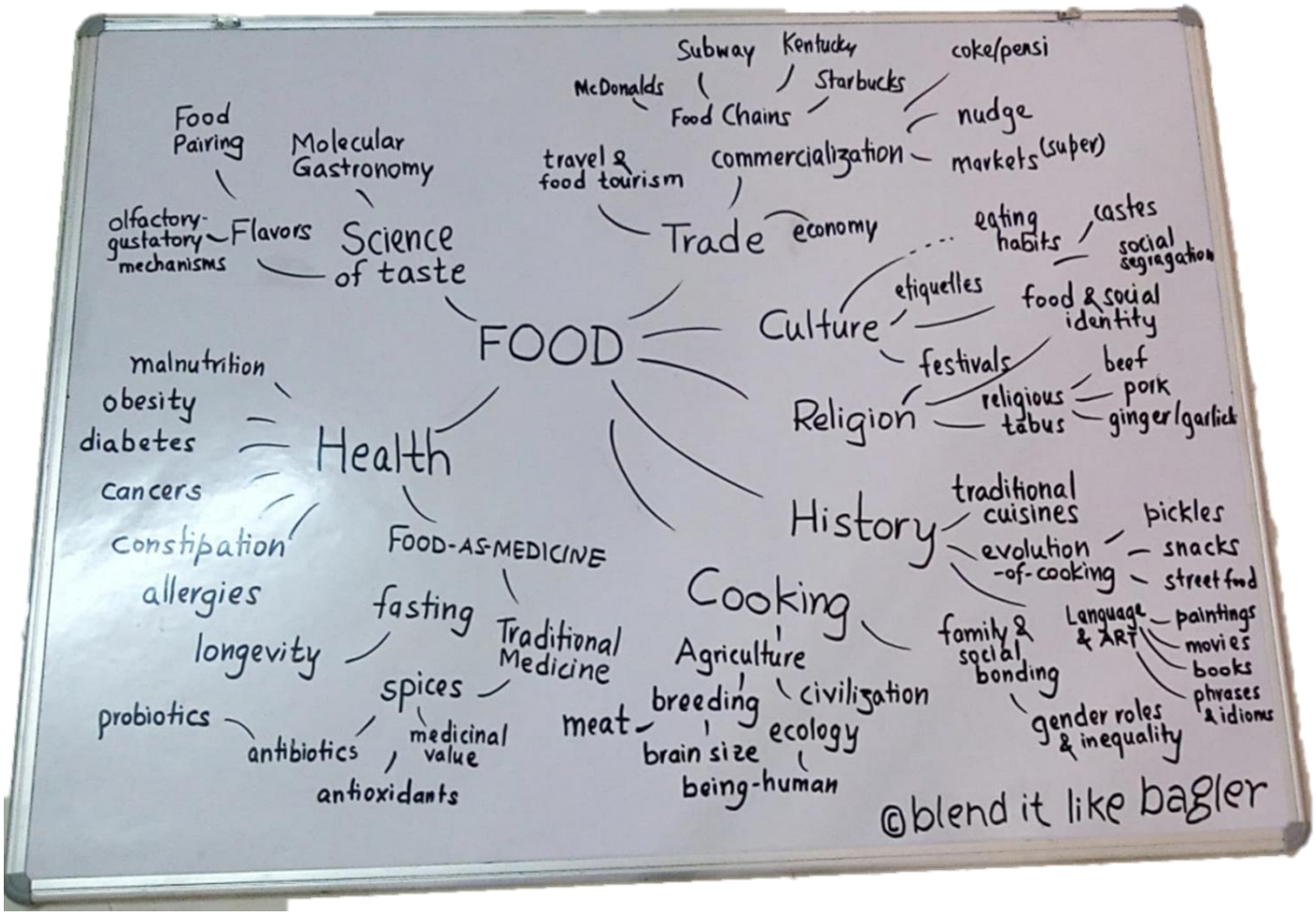
- 'The birth of modern diet' by Rachel Laudan, Scientific American, pages 64-71, Scientific American, The Science of Food Issue, Summer 2015.
  - This article is provided separately.

- 'The amazing multimillion-year history of processed food' by Rachel Laudan, Scientific American, pages 78-83, Scientific American, The Science of Food Issue, Summer 2015.
  - This article is provided separately.

- **Book Suggestions for Future Reading:**
  - Cooked by Michael Pollan
  - The Omnivor's Dilemma by Michael Pollan
  - Fast Food Nation by Eric Schlossler

- **Genetically Modified Food:** Arguments for and against it. (BT Brinjal)

- **Various dimensions of food:** Food is central to the cultural identity of humans.



COOKING: Being human; Exceptionally large brain size of *Homo sapiens*; Agriculture-Civilization; Breeding; Meat; Family and social bonding; Gender roles and inequality.

HISTORY: Evolution of cooking; Traditional cuisines across the world; Recipes; Pickles, snacks, and street food; Languages and Art: paintings, movies, books, phrases, and idioms.

RELIGION & CULTURE: Religious taboos: beef, pork, ginger/garlic; Food and social identity; Festivals; Etiquettes; Food habits; Castes and communities; Food-based social segregation.

TRADE: Economy of the food industry; Travel & Tourism; Food processing and commercialization of food; Supermarkets; Food chains: McDonald's, Kentucky fried chicken, Starbucks, Subway, Coke, Pepsi; Nudging and choice design.

SCIENCE: Food micronutrients & macronutrients; Food processing and preservation; Preservatives; Flavors; Food pairing; Neural mechanisms; Neuro-gastronomy; Flavor molecules; Olfactory and gustatory mechanisms; Molecular gastronomy; Computational gastronomy; Data-driven food innovations.

HEALTH: Malnutrition; Lifestyle disorders; Non-communicable diseases; Obesity; Diabetes, Cardiovascular disorders; Irritable bowel syndrome; Crohn's disease; Cancers; Constipation; Allergies; Gut microbiome; Food-As-A-Medicine: Ayurveda, Traditional medicinal systems; Ethno-medicine; Fasting; Longevity; Culinary herbs & spices and their medicinal value; Antioxidants; Probiotics, Antibiotics.